

Sri Lanka Institute of Information technology

BSc (Hons) in Information Technology - 2026



IE3132-Penetration Testing

Cyber Security - Year 3, Semester 1

Assignment 01

Assignment Title	CTF Play Box Implementation-Project Initiation
Learning outcomes covered	LO1:Apply information gathering techniques in the penetration testing process LO2:Evaluate the necessary penetration testing tools and techniques in a given scenario LO3: Develop exploitation code to facilitate penetration testing
Assignment Mode	Individual
Maximum Marks	100
Contribution to the Final Grade	10%
Date Published	11/08/2026
Deadline for submissions	25/09/2026
Mode of Submission	Electronic submission of the Report and viva based on the report. (http://courseweb.sliit.lk)

Assignment Description:

This individual assignment represents the initial stage of a Capture The Flag (CTF) Play Box implementation. The purpose of this stage is to design and document the proposed CTF platform, challenge categories, box architecture, challenge flow, and individual team responsibilities. Students are not required to deliver the complete final CTF system at this stage. The submitted design must, however, be technically realistic and sufficiently detailed to support later implementation.

Assignment Objectives

- Identify suitable cybersecurity domains and convert them into a structured CTF challenge pathway.
- Design a coherent CTF Play Box containing multiple progressive stages with clear learning outcomes.
- Define the proposed platform, box architecture, services, operating environment, and deployment approach.

- Demonstrate understanding of practical cybersecurity concepts through challenge design.
- Plan the technical implementation and testing requirements before the full CTF build.
- Clearly document the contribution and responsibility of each member of the four-person group.

Minimum CTF Stage Requirements

Each group shall propose a minimum of 6 stages/challenges. The six stages must cover at least 4 different cybersecurity domains. Groups may propose additional stages where technically justified. The final implementation in later stages must follow the approved design unless changes are documented.

No.	Recommended Domain / Field	Example Challenge Focus
1	Steganography	Hidden data in images, audio, metadata, files
2	Web Technologies / Web Security	Web vulnerabilities, authentication, input validation
3	Cryptography	Encoding, classical ciphers, weak cryptographic implementation
4	Digital Forensics	File analysis, logs, deleted/hidden artefacts
5	Networking	Packet analysis, protocols, network enumeration
6	OSINT / Reconnaissance	Public information discovery and correlation
7	Reverse Engineering	Binary/application analysis and basic program logic
8	Linux / System Security	Permissions, services, command-line investigation
9	Programming / Scripting	Logic-based challenges, scripting, automation
10	Miscellaneous	Puzzles or hybrid challenges linked to cybersecurity

Note: The list above is a recommended domain list, not a requirement to use every domain. The group must justify why its selected domains are appropriate for the intended target participants.

Expected Difficulty Progression

Stage	Suggested Difficulty	Expected Purpose
1–2	Easy	Introduce the scenario and basic concepts
3–4	Moderate	Require analysis and application of security techniques
5	Moderate–Hard	Combine multiple concepts or investigation steps

6	Hard	Provide an integrating or capstone-style challenge
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Basic Information of the Proposed CTF Platform

The assignment report must define the following information for the proposed CTF Play Box:

Required Field	Information to Provide
CTF / Box Name	Unique name and short description
Theme / Scenario	Storyline or security scenario connecting the challenges
Target Audience	Expected skill level and intended participant group
Learning Outcomes	Technical skills participants should gain
Challenge Count	Minimum 6 proposed stages; additional stages may be included
Challenge Domains	Selected cybersecurity fields and justification
Difficulty Model	Easy / Moderate / Hard progression
Flag Format	Proposed flag naming and validation format
Platform Type	Web-based, VM-based, container-based, hybrid, etc.
Operating Environment	OS, versions, architecture, and required packages
Deployment Method	Local server, virtual machine, Docker, cloud, or hybrid
Network Design	Required ports, services, isolation, and connectivity
Security Controls	Isolation, reset mechanism, privilege restrictions, logging
Tools / Technologies	Programming languages, frameworks, databases, tools
Estimated Resources	CPU, RAM, storage, network, and software requirements

Challenge / Stage Specification

For each proposed stage, provide a separate entry containing the following:

Field	Required Description
Stage ID & Title	Unique identifier and meaningful challenge title
Domain	e.g., Web, Steganography, Forensics, Crypto
Difficulty	Easy, Moderate, or Hard with justification
Scenario	Short realistic problem statement
Learning Objective	Specific skill or concept being assessed
Challenge Environment	VM/container/web application/file set/etc.
Player Task	What the participant is expected to discover or achieve

Flag Location / Logic	How the flag is obtained; do not reveal unnecessary answers
Required Tools	Expected tools, commands, or techniques
Dependencies	Any prerequisite stage, service, file, or knowledge
Expected Solution Path	High-level intended solving approach
Validation	How successful completion will be verified
Hints	Optional progressive hints and proposed penalties
Reset / Recovery	How the challenge can be restored to its initial state

Platform and Box Architecture

Include a simple architecture diagram showing the participant machine, CTF platform/control layer, individual challenge boxes or containers, network segmentation, flag validation mechanism, and any supporting services. The diagram must identify major technologies and communication paths.

Required Deliverables for Initial Stage

- Assignment report / design document.
- CTF platform and box architecture diagram.
- Minimum six-stage challenge plan covering at least four cybersecurity domains.
- Detailed specification for every proposed stage.
- Proposed technology stack and deployment plan.
- Four-member individual contribution matrix.
- Risk, dependency, and implementation considerations.
- Short group presentation or design walkthrough, if required by the lecturer.

Group Member Roles and Responsibilities:

Although the assignment is completed as a group, each member must have a clearly identifiable technical responsibility. The following allocation is recommended; groups may modify it provided that responsibilities are balanced and approved.

Member	Suggested Responsibility	Expected Evidence
Member 1	CTF Platform & Architecture	Architecture diagram, platform design, deployment plan, environment specification
Member 2	Challenge Design A	At least 3 stage specifications, challenge logic, tools, flags, hints
Member 3	Challenge Design B	At least 3 stage specifications, challenge logic, tools, flags, hints
Member 4	Integration, Testing & Documentation	Testing strategy, validation, risk analysis, documentation integration

Note: All team members are expected to contribute across tasks beyond their role to ensure cohesion.

Usage of AI: For the completion of assignment, students may use a freely available generative AI tool. However, you need to be careful with how you use the prompt in order to build up a suitable scenario for the assignment. In case generative AI is used for scenario build up, student's shall include the prompt log in the Annexures of the report.

Marking Scheme

Recommended assessment weight: 100 marks for this assignment, contributing 10% to the overall module assessment.

Assessment Criterion	Excellent	Good	Satisfactory	Poor / Incomplete	Marks
CTF concept & stage planning	Clear, coherent and technically justified design	Mostly coherent with minor gaps	Basic planning with limited justification	Unclear or incomplete	15
Domain coverage & challenge quality	≥6 stages, ≥4 domains; strong technical relevance	Requirements met with minor weaknesses	Minimum requirements barely met	Requirements not met	15
Stage specifications	Detailed, realistic, solvable and well structured	Detailed with some missing elements	Basic specifications	Major information missing	20

Platform & box architecture	Robust architecture, deployment and isolation plan	Good architecture with minor gaps	Basic architecture	Unclear/infeasible	15
Technical feasibility & security	Strong technology choices, dependencies, risks and isolation	Mostly feasible and secure	Some technical gaps	Major feasibility/security issues	10
Learning outcomes & difficulty progression	Excellent alignment and progression	Good alignment	Limited alignment	Poor/no alignment	5
Documentation & presentation	Professional, precise and easy to follow	Well presented with minor issues	Acceptable documentation	Poorly organized	5
Individual contribution	Balanced, evidenced and technically substantial	Clear contributions with adequate evidence	Contributions identified but limited evidence	Unclear/unequal contribution	10
Testing / validation plan	Comprehensive validation and recovery approach	Good testing plan	Basic testing plan	Minimal/no testing plan	5
Total					100

Assessment Expectations

- Technical depth is expected; simply listing challenge titles is insufficient.
- Each challenge must have a clear objective, intended solving path, environment, flag mechanism, and validation method.
- The proposed environment must be isolated and designed to prevent unintended impact on institutional or public systems.
- Use only intentionally vulnerable systems, datasets, applications, and configurations created or authorized for the CTF.
- All external tools, frameworks, datasets, or references used in the design should be acknowledged.
- The initial-stage assignment is a design and planning milestone; students should not claim completed implementation unless it has actually been developed and tested.

Suggested Report Structure

Cover Page → Executive Summary → Objectives → CTF Concept & Scenario → Selected Domains → Stage Specifications → Platform/Box Architecture → Technology Stack → Deployment & Isolation Plan → Testing/Validation Plan → Individual Contributions → Risks/Limitations → Conclusion → References → Appendices.

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